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- Utilizando la serie de tiempo `jj` del paquete `astsa` (ganancias trimestrales de Johnson & Johnson, 1960–1980), desarrolle una aplicación en Shiny que permita seleccionar y comparar métodos de suavizamiento: Medias Móviles, SES y Holt-Winters. La interfaz dispondrá de un panel lateral con un control deslizante (`sliderInput`) para ajustar el horizonte de pronóstico (1 a 12 trimestres) y un panel principal de dos pestañas: la primera mostrará la descomposición de la serie (tendencia, estacionalidad y residuo) y la segunda, el gráfico del pronóstico con sus intervalos de confianza.

- Generate  $n = 100$  observations from the autoregression

$$x_t = -0.9x_{t-2} + w_t$$

with  $\sigma_w = 1$ . Next, apply the moving average filter

$$v_t = (x_t + x_{t-1} + x_{t-2} + x_{t-3})/4,$$

to  $x_t$ , the data you generated. Now plot  $x_t$  as a line and superimpose  $v_t$  as a dashed line. Comment on the behavior of  $x_t$  and how applying the moving average filter changes that behavior. (Hints: Use `filter` function the R)

- Repeat (a) but with

$$x_t = \cos(2\pi t/4).$$

- Repeat (b) but with added  $N(0, 1)$  noise,

$$x_t = \cos(2\pi t/4) + w_t.$$

- Compare and contrast (a)-(c); i.e., how does the moving average change each series.

- Consider the time series

$$x_t = \beta_1 + \beta_2 t + w_t,$$

where  $\beta_1$  and  $\beta_2$  are known constants and  $w_t$  is a white noise process with variance  $\sigma_w^2$ .

- Determine whether  $x_t$  is stationary.
- Show that the process  $y_t = x_t - x_{t-1}$  is stationary.
- Show that the mean of the moving average

$$v_t = \frac{1}{2n+1} \sum_{j=-n}^n x_{t-j}, \quad (1)$$

is  $\beta_1 + \beta_2 t$ , and give a simplified expression for the autocovariance function.

- For a moving average process of the form

$$x_t = w_{t-1} + 2w_t + w_{t+1},$$

where  $w_t$  are independent with zero means and variance  $\sigma_w^2$ , determine the autocovariance and autocorrelation functions as a function of lag  $h = s - t$  and plot the ACF as a function of  $h$ .

- Consider the random walk with drift model

$$x_t = \delta + x_{t-1} + w_t,$$

for  $t \in \mathbb{N}$  with  $x_0 = 0$ , where  $w_t$  is white noise with variance  $\sigma_w^2$ .

- Show that the model can be written as  $x_t = \delta t + \sum_{k=1}^t w_k$ .
- Find the mean function and the autocovariance function of  $x_t$ .
- Argue that  $x_t$  is not stationary.

- d. Show  $\rho_x(t-1, t) = \sqrt{\frac{t-1}{t}} \rightarrow 1$  as  $t \rightarrow \infty$ . What is the implication of this result?
- e. Suggest a transformation to make the series stationary, and prove that the transformed series is stationary.
6. For an  $MA(1)$ ,  $x_t = w_t + \theta w_{t-1}$ , show that  $|\rho_x(1)| \leq 1/2$  for any number  $\theta$ . For which values of  $\theta$  does  $\rho_x(1)$  attain its maximum and minimum?
7. A real-valued function  $g(t)$ , defined on the integers, is non-negative definite if and only if

$$\sum_{i=1}^n \sum_{j=1}^n a_i g(t_i - t_j) a_j \geq 0,$$

for all positive integers  $n$  and for all vectors  $a = (a_1, a_2, \dots, a_n)'$  and  $t = (t_1, t_2, \dots, t_n)'$ . For the matrix  $G = \{g(t_i - t_j); i, j = 1, 2, \dots, n\}$ , this implies that  $a'Ga \geq 0$  for all vectors  $a$ .

- Prove that  $\gamma(h)$ , the autocovariance function of a stationary process, is a non-negative definite function.
  - Verify that the sample autocovariance  $\hat{\gamma}(h)$ , is a non-negative definite function.
8. Suppose we would like to predict a single stationary series  $x_t$  with zero mean and autocorrelation function  $\gamma(h)$  at some time in the future, say,  $t + l$ , for  $l > 0$ .
- If we predict using only  $x_t$  and some scale multiplier  $A$ , show that the mean-square prediction error

$$MSE(A) = \mathbb{E}[(x_{t+l} - Ax_t)^2]$$

is minimized by the value

$$A = \rho(l).$$

- Show that the minimum mean-square prediction error is

$$MSE(A) = \gamma(0)[1 - \rho^2(l)].$$

- Show that if  $x_{t+l} = Ax_t$ , then  $\rho(l) = 1$  if  $A > 0$ , and  $\rho(l) = -1$  if  $A < 0$ .

9. Sea  $y_t = a_t + ca_{t-1} + ca_{t-2} + \dots + ca_1$ , para  $t > 0$ , donde  $c \in \mathbb{R}$  y  $a_t \sim RB(0, \sigma_a^2)$ .

- Calcular la media y autocovarianza de  $y_t$ . ¿Es estacionaria?
- Demostrar que la serie  $z_t = y_t - y_{t-1}$  es estacionaria.

10. Consider the two series

$$x_t = w_t$$

$$y_t = w_t - \theta w_{t-1} + u_t,$$

where  $w_t$  and  $u_t$  are independent white noise series with variances  $\sigma_w^2$  and  $\sigma_u^2$ , respectively, and  $\theta$  is an unspecified constant.

- Express the ACF,  $\rho_y(h)$ , for  $h \in \mathbb{Z}$  of the series  $y_t$  as a function of  $\sigma_w^2$ ,  $\sigma_u^2$ , and  $\theta$ .
- Determine the CCF,  $\rho_{xy}(h)$  relating  $x_t$  and  $y_t$ .
- Show that  $x_t$  and  $y_t$  are jointly stationary.

11. A concept used in geostatistics, see Journel and Huijbregts, is that of the variogram, defined for a spatial process  $x_s$ ,  $s = (s_1, s_2)$ , for  $s_1, s_2 \in \mathbb{Z}$  as

$$V_x(h) = \frac{1}{2} \mathbb{E}[(x_{s+h} - x_s)^2],$$

where  $h = (h_1, h_2)$ , for  $h_1, h_2 \in \mathbb{Z}$ . Show that, for a stationary process, the variogram and autocovariance functions can be related through

$$V_x(h) = \gamma(0) - \gamma(h),$$

where  $\gamma(h)$  is the usual lag  $h$  covariance function and  $0 = (0, 0)$ . Note the easy extension to any spatial dimension.

12. Consider the series

$$x_t = \sin(2\pi Ut),$$

$t \in \mathbb{N}$ , where  $U$  has a uniform distribution on the interval  $(0, 1)$ .

- a. Prove  $x_t$  is weakly stationary.
  - b. Prove  $x_t$  is not strictly stationary.
13. Show that a strictly stationary process with  $\mathbb{E}(X_t^2) < \infty$  is weakly stationary.
14. Let  $x_t$  be a stationary normal process with mean  $\mu_x$  and autocovariance function  $\gamma(h)$ . Define the nonlinear time series

$$y_t = \exp x_t.$$

- a. Express the mean function  $\mathbb{E}[y_t]$  in terms of  $\mu_x$  and  $\gamma(0)$ . The moment generating function of a normal random variable  $x$  with mean  $\mu$  and variance  $\sigma^2$  is
 
$$M_x(\lambda) = \mathbb{E}[\exp\{\lambda x\}] = \exp\left\{\mu\lambda + \frac{1}{2}\sigma^2\lambda^2\right\}.$$
  - b. Determine the autocovariance function of  $y_t$ . The sum of the two normal random variables  $x_{t+h} + x_t$  is still a normal random variable.
15. Let  $\{z_t\}$  be a sequence of independent normal random variables, each with mean 0 and variance  $\sigma^2$ , and let  $a$ ,  $b$ , and  $c$  be constants. Which, if any, of the following processes are stationary? For each stationary process specify the mean and autocovariance function.
- a.  $x_t = a + bz_t + cz_{t-2}$
  - b.  $x_t = z_1 \cos(ct) + z_2 \sin(ct)$
  - c.  $x_t = z_t \cos(ct) + z_{t-1} \sin(ct)$
  - d.  $x_t = a + bz_0$
  - e.  $x_t = z_0 \cos(ct)$
  - f.  $x_t = z_t z_{t-1}$

16. Let  $\{x_t\}$  be the moving-average process of order 2 given by

$$x_t = z_t + \theta z_{t-2},$$

where  $\{z_t\}$  is  $\text{WN}(0, 1)$ .

- a. Find the autocovariance and autocorrelation functions for this process when  $\theta = 0,8$ .
  - b. Compute the variance of the sample mean  $(x_1 + x_2 + x_3 + x_4)/4$  when  $\theta = 0,8$ .
  - c. Repeat (b) when  $\theta = -0,8$  and compare your answer with the result obtained in (b).
17. Let  $w_t$ , for  $t \in \mathbb{N}$  be a normal white noise process, and consider the series

$$x_t = w_t w_{t-1}.$$

Determine the mean and autocovariance function of  $x_t$ , and state whether it is stationary.

18. a. Simulate a series of  $n = 500$  moving average observations as in exercise 3 and compute the sample ACF,  $\hat{\rho}(h)$ , to lag 20. Compare the sample ACF you obtain to the actual ACF,  $\rho(h)$ .
  - b. Repeat part (a) using only  $n = 50$ . How does changing  $n$  affect the results?
19. For the AR series shown below, determine a set of difference equations that can be used to find the ACF  $\rho(h)$ ,  $h = 0, 1, \dots$ ; solve for the constants in the ACF using the initial conditions. Then plot the ACF values to lag 10 (use `ARMAacf` as a check on your answers).
- a.  $x_t + 1,6x_{t-1} + ,64x_{t-2} = w_t$
  - b.  $x_t - ,40x_{t-1} - ,45x_{t-2} = w_t$
  - c.  $x_t - 1,2x_{t-1} + ,85x_{t-2} = w_t$
20. Simulate a series of 200 observations from each of the models with  $\sigma_w^2 = 1$  in Exercise 15. For each case, plot the simulate series, and calculate and study its sample ACF  $\hat{\rho}(h)$  and PACF  $\hat{\phi}_{hh}$  for  $h = 1, \dots, 20$ .

21. Show that if an AR(2) process is stationary, then

$$\rho^2(1) < \frac{\rho(2) + 1}{2}.$$

22. Find the range of  $\alpha$  such that the AR(2) process

$$x_t = x_{t-1} + \alpha x_{t-2} + w_t$$

is stationary. Determine the ACF for  $\alpha = -1/2$ .

23. ¿Cuáles de las siguientes funciones definidas en  $\mathbb{Z}$  son funciones de autocovarianza de un proceso estacionario?

a.  $\gamma(h) = \begin{cases} 1, & \text{si } h = 0; \\ \frac{1}{h}, & \text{si } h \neq 0. \end{cases}$

b.  $\gamma(h) = (-1)^{|h|}$

c.  $\gamma(h) = 1 + \cos\left(\frac{\pi h}{2}\right) + \cos\left(\frac{\pi h}{4}\right)$

d.  $\gamma(h) = 1 + \cos\left(\frac{\pi h}{2}\right) - \cos\left(\frac{\pi h}{4}\right)$

e.  $\gamma(h) = \begin{cases} 1, & \text{si } h = 0; \\ 0,4, & \text{si } h = \pm 1; \\ 0, & \text{si } h \neq 0, \pm 1. \end{cases}$

24. Find a process that has the following autocovariance function:

$$\gamma(0) = 10, \quad \gamma(1) = 0, \quad \gamma(2) = -4, \quad \gamma(h) = 0, \quad \text{for } |h| > 2.$$

Examine stationarity and invertibility for the process obtained.

25. Let  $\{w_t; t = 0, 1, \dots\}$  be a white noise process with variance  $\sigma_w^2$  and let  $|\phi| < 1$  be a constant. Consider the process  $x_0 = w_0$ , and

$$x_t = \phi x_{t-1} + w_t, \quad t = 1, 2, \dots$$

We might use this method to simulate an AR(1) process from simulated white noise.

a. Show that  $x_t = \sum_{j=0}^t \phi^j w_{t-j}$  for any  $t = 0, 1, \dots$

b. Find the  $\mathbb{E}(x_t)$ .

c. Show that, for  $t = 0, 1, \dots$ ,

$$\text{Var}(x_t) = \frac{\sigma_w^2}{1 - \phi^2} (1 - \phi^{2(t+1)})$$

d. Show that, for  $h \geq 0$ ,

$$\text{Cov}(x_{t+h}, x_t) = \phi^h \text{Var}(x_t)$$

e. Is  $x_t$  stationary?

f. Argue that, as  $t \rightarrow \infty$ , the process becomes stationary, so in a sense,  $x_t$  is "asymptotically stationary."

g. Comment on how you could use these results to simulate  $n$  observations of a stationary Gaussian AR(1) model from simulated iid  $N(0, 1)$  values.

h. Now suppose  $x_0 = w_0/\sqrt{1 - \phi^2}$ . Is this process stationary?

26. Suponga que  $x_t = m_t + z_t$ , donde  $\{z_t, t \in \mathbb{Z}\}$  es una secuencia de variables aleatorias normales independientes con medio cero y varianza  $\sigma^2$ . Además

$$m_t = c_0 + c_1 t + c_2 t^2,$$

con  $c_0, c_1$  y  $c_2$  constantes. Definamos los siguientes procesos:

$$u_t = \sum_{i=-2}^2 a_i x_{t+i}; \quad a_2 = a_{-2} = -3/35, \quad a_1 = a_{-1} = 12/35, \quad a_0 = 17/35.$$

$$w_t = \sum_{i=-3}^3 b_i x_{t+i}; \quad b_3 = b_{-3} = -2/21, \quad b_2 = b_{-2} = 3/21, \quad b_1 = b_{-1} = 6/21, \quad b_0 = 7/21.$$

a. Encontrar la media y varianza de  $u_t$  y  $w_t$ .

b. Calcular  $\rho(u_t, u_{t+1})$  y  $\rho(w_t, w_{t+1})$ .

c. Determinar  $\rho_{uw}(u_t, w_{t+1})$ .

27. Sea  $z_t$  un ruido iid  $N(0, 1)$  y defina:

$$x_t = \begin{cases} z_t, & \text{si } t \text{ es par,} \\ (z_{t-1}^2 - 1)/\sqrt{2}, & \text{si } t \text{ es impar.} \end{cases}$$

a. Demostrar que  $x_t$  es  $RB(0, 1)$ , pero no es un ruido  $iid(0, 1)$ .

b. Hallar  $\mathbb{E}[x_{n+1}|x_1, x_2, \dots, x_n]$ , para  $n$  par e impar y compare los resultados.

28. Demostrar que el valor de la PACF de rezago 2 en una proceso MA(1)

$$x_t = z_t + \theta z_{t-1}, \quad t \in \mathbb{Z},$$

donde  $\{z_t\}$  es un ruido blanco con media cero y varianza  $\sigma^2$  es

$$\phi_{22} = -\theta^2/(1 + \theta^2 + \theta^4)$$

29. In this problem **2.9**, we will explore the periodic nature of  $S_t$ , the SOI series displayed in *Figure 1.5*.

a. Detrend the series by fitting a regression of  $S_t$  on time  $t$ . Is there a significant trend in the sea surface temperature? Comment.

b. Calculate the periodogram for the detrended series obtained in part (a). Identify the frequencies of the two main peaks (with an obvious one at the frequency of one cycle every 12 months). What is the probable El Niño cycle indicated by the minor peak?

30. Pruebe que  $\Delta^r z_t = \sum_{j=0}^r (-1)^j \binom{r}{j} z_{t-j}$ .

31. Suponga que  $\{a_t, t = 1, 2, \dots\}$  es una secuencia de v.a. independientes e idénticamente distribuidas, con:

$$P(a_t = 0) = P(a_t = 1) = \frac{1}{2}.$$

a. ¿Es estacionario el proceso  $a_1 + a_2 \cos t$ ?

b. ¿Es estacionario el proceso  $a_1 + a_2 \cos t + a_3 \cos t + \sin t$ ?

32. Si  $\{x_t, t \in T\}$  y  $\{y_t, t \in T\}$  son estacionarios e independientes, ¿será estacionario el proceso  $\{ax_t + by_t, t \in T\}$ ?

33. Se lanza una moneda justa para generar  $x_t$ , donde  $x_t = 1$  si cae cara y  $x_t = 0$  si cae sello. Se define la serie:

$$y_t = 5 + x_t - 0,65 x_{t-1}$$

a. Comparar la función de autocorrelación muestral (FAC) con la función de autocorrelación teórica  $\rho(h)$  para distintos tamaños muestrales  $n = 10, 100, 200, 500, 1000$ .

b. Para cada tamaño  $n$ , simular 1000 réplicas, calcular la FAC muestral hasta el rezago 10 y verificar la distribución muestral de la FAC.

34. For a linear process of the form

$$x_t = \sum_{j=0}^{\infty} \phi^j w_{t-j},$$

where  $\{w_t\}$  satisfies the conditions of Theorem A.7 and  $|\phi| < 1$ , show that

$$\sqrt{n} \frac{(\hat{\rho}_x(1) - \rho_x(1))}{\sqrt{1 - \rho_x^2(1)}} \xrightarrow{d} N(0, 1),$$

and construct a 95 % confidence interval for  $\phi$  when  $\hat{\rho}_x(1) = 0,64$  and  $n = 100$ .